

Отчет

Лабораторная работа 5.1. Развертывание и настройка Hadoop Анализ данных с использованием экосистемы Hadoop

Цель работы: получить практические навыки развертывания одноузлового кластера Hadoop, освоить базовые операции с распределенной файловой системой HDFS, выполнить загрузку и простейшую обработку данных, а также научиться выгружать результаты для последующего анализа и визуализации во внешней среде (Jupyter Notebook / Google Colab).

- 1) напишите Python-скрипт для создания таблицы в PostgreSQL и загрузки в нее данных из вашего CSV-файла. это имитирует наличие данных в операционной реляционной СУБД.

напишите Python-скрипт для создания таблицы в PostgreSQL и загрузки в нее данных из вашего CSV-файла. это имитирует наличие данных в операционной реляционной СУБД.

```
: import pandas as pd
from sqlalchemy import create_engine

DB_USER = "postgres"
DB_PASS = "123"
DB_NAME = "lab5_db"
DB_HOST = "localhost"
DB_PORT = "5432"

engine = create_engine(f"postgresql://{DB_USER}:{DB_PASS}@{DB_HOST}:{DB_PORT}/{DB_NAME}")

csv_path = "/Users/sofia/Desktop/AB_NYC_2019.csv"
df = pd.read_csv(csv_path)

print("Строк:", len(df))

df.to_sql("airbnb", engine, if_exists="replace", index=False)

print("Данные успешно загружены!")
```

Строк: 48895
Данные успешно загружены!

Вариант 17

Аренда жилья Airbnb Нью-Йорк

Средняя цена и количество предложений по району (Hive)

Выполните аналитическую задачу из вашего варианта, написав простые скрипты на Bash (например, с использованием grep, awk, sort, uniq) или Python, которые будут выполнять роль mapper и reducer.

```
Last login: Wed Nov 26 08:58:47 on ttys000
You have new mail
(base) sofia@MacBook-Air-2 ~ % cd /usr/local/hadoop

stop-dfs.sh
stop-yarn.sh

start-dfs.sh
start-yarn.sh

jps

Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [MacBook-Air-2.local]
2025-11-26 08:59:21,245 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Stopping nodemanagers
Stopping resourcemanager
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [MacBook-Air-2.local]
2025-11-26 08:59:34,813 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting resourcemanager
Starting nodemanagers
34358 NodeManager
34439 Jps
33897 DataNode
33786 NameNode
34044 SecondaryNameNode
34253 ResourceManager
(base) sofia@MacBook-Air-2 ~ % hadoop % hdfs dfs -ls /user/sofia/lab5

2025-11-26 08:59:52,666 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 4 items
-rw-r--r-- 1 sofia supergroup 7077973 2025-11-24 22:25 /user/sofia/lab5/input.csv
-rw-r--r-- 1 sofia supergroup 498 2025-11-24 22:36 /user/sofia/lab5/mapper.py
drwxr-xr-x - sofia supergroup 0 2025-11-24 22:54 /user/sofia/lab5/output_price_by_area
-rw-r--r-- 1 sofia supergroup 679 2025-11-24 22:36 /user/sofia/lab5/reducer.py
(base) sofia@MacBook-Air-2 ~ % hadoop % hdfs dfs -rm -r /user/sofia/lab5/output_price_by_area
2025-11-26 09:00:00,581 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Deleted /user/sofia/lab5/output_price_by_area
(base) sofia@MacBook-Air-2 ~ % hadoop % cd ~
ls -l mapper.py reducer.py

-rwxr-xr-x 1 sofia staff 498 24 Nov 22:36 mapper.py
-rwxr-xr-x 1 sofia staff 679 24 Nov 22:36 reducer.py
(base) sofia@MacBook-Air-2 ~ % hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
-input /user/sofia/lab5/input.csv \
-output /user/sofia/lab5/output_price_by_area \
-mapper mapper.py \
-reducer reducer.py \
-file mapper.py \
-file reducer.py

2025-11-26 09:00:14,371 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
2025-11-26 09:00:14,499 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
packageJobJar: [mapper.py, reducer.py, /var/folders/wf/q9h5vr9x74gd9d5q68_1m4m0000gn/T/hadoop-unjar3539241962237168398/] [] /var/folders/wf/q9h5vr9x74gd9d5q68_1m4m0000gn/T/streamjob11955312018540567659
jar tmpDm1null
2025-11-26 09:00:15,051 INFO client.DefaultHohHARMAFailoverProxyProvider: Connecting to ResourceManager at localhost/127.0.0.1:8032
2025-11-26 09:00:15,181 INFO client.DefaultHohHARMAFailoverProxyProvider: Connecting to ResourceManager at localhost/127.0.0.1:8032
2025-11-26 09:00:15,374 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/sofia/.staging/job_1764136778124_0001

2025-11-26 09:00:17,086 INFO mapred.FileInputFormat: Total input files to process : 1
2025-11-26 09:00:18,019 INFO mapreduce.JobSubmitter: number of splits:2
2025-11-26 09:00:18,585 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1764136778124_0001
2025-11-26 09:00:18,585 INFO mapreduce.JobSubmitter: Executing with tokens: []
2025-11-26 09:00:18,721 INFO conf.Configuration: resource-types.xml not found
2025-11-26 09:00:19,721 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2025-11-26 09:00:19,062 INFO impl.VarClientImpl: Submitted application application_1764136778124_0001
2025-11-26 09:00:19,084 INFO mapreduce.Job: The url to track the job: http://localhost:8080/proxy/application_1764136778124_0001/
2025-11-26 09:00:19,085 INFO mapreduce.Job: Running job: job_1764136778124_0001
2025-11-26 09:00:25,215 INFO mapreduce.Job: Job job_1764136778124_0001 running in uber mode : false
2025-11-26 09:00:25,223 INFO mapreduce.Job: map 0% reduce 0%
2025-11-26 09:00:33,380 INFO mapreduce.Job: map 100% reduce 0%
2025-11-26 09:00:38,425 INFO mapreduce.Job: map 100% reduce 100%
2025-11-26 09:00:39,452 INFO mapreduce.Job: Job job_1764136778124_0001 completed successfully
2025-11-26 09:00:39,548 INFO mapreduce.Job: Counters: 50
File System Counters
FILE: Number of bytes read=818539
FILE: Number of bytes written=2477956
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=7052267
HDFS: Number of bytes written=195
HDFS: Number of read operations=11
HDFS: Number of large read operations=0
HDFS: Number of write operations=2
HDFS: Number of bytes read erasure-coded=0
Job Counters
Launched map tasks=2
Launched reduce tasks=1
Data-local map tasks=2
Total time spent by all maps in occupied slots (ms)=12820
Total time spent by all reduces in occupied slots (ms)=2234
Total time spent by all map tasks (ms)=12820
Total time spent by all reduce tasks (ms)=2234
Total vcore-milliseconds taken by all map tasks=12820
Total vcore-milliseconds taken by all reduce tasks=2234
Total megabyte-milliseconds taken by all map tasks=13127680
Total megabyte-milliseconds taken by all reduce tasks=2287616
Map-Reduce Framework
Map input records=49081
Map output records=48894
Map output bytes=720745
Map output materialized bytes=818545
Input split bytes=198
Combine input records=0
Combine output records=0
Reduce input groups=5
Reduce shuffle bytes=818545
Reduce input records=48894
Reduce output records=5
Spilled Records=97708
Shuffled Maps =2
Failed Shuffles=0
Merged Map Outputs=2
GC time elapsed (ms)=55
CPU time spent (ms)=0
Physical memory (bytes) snapshot=0
Virtual memory (bytes) snapshot=0
Total committed heap usage (bytes)=414187520
Shuffle Errors
BAD_ID=0
CONNECTION=0
```

```
FILE: Number of bytes written=2477956
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=7082267
HDFS: Number of bytes written=105
HDFS: Number of read operations=11
HDFS: Number of large read operations=0
HDFS: Number of write operations=2
HDFS: Number of bytes read erasure-coded=0

Job Counters
  Launched map tasks=2
    Launched reduce tasks=1
  Data-local map tasks=2
  Total time spent by all maps in occupied slots (ms)=12820
  Total time spent by all reduces in occupied slots (ms)=2234
  Total time spent by all map tasks (ms)=12820
  Total time spent by all reduce tasks (ms)=2234
  Total vcore-milliseconds taken by all map tasks=12820
  Total vcore-milliseconds taken by all reduce tasks=2234
  Total megabyte-milliseconds taken by all map tasks=13127600
  Total megabyte-milliseconds taken by all reduce tasks=2287616

Map-Reduce Framework
  Map input records=49081
  Map output records=48894
  Map output bytes=728745
  Map output materialized bytes=818545
  Input split bytes=198
  Combine input records=0
  Combine output records=0
  Reduce input groups=5
  Reduce shuffle bytes=818545
  Reduce input records=48894
  Reduce output records=5
  Spilled Records=97788
  Shuffled Maps =2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=55
  CPU time spent (ms)=0
  Physical memory (bytes) snapshot=0
  Virtual memory (bytes) snapshot=0
  Total committed heap usage (bytes)=414187520

Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=7082069

File Output Format Counters
  Bytes Written=105

2025-11-26 09:00:39.548 INFO streaming.StreamJob: Output directory: /user/sofia/lab5/output_price_by_area
(base) sofia@MacBook-Air-2 ~ % hdfs dfs -cat /user/sofia/lab5/output_price_by_area/part-00000
2025-11-26 09:00:47.481 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Bronx      87.50      1091
Brooklyn   124.39    20103
Manhattan  196.88    21661
Queens     99.52     5666
Staten Island 114.81    373
(base) sofia@MacBook-Air-2 ~ % █
```

Часть 3. Анализ и визуализация

```

import os
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Стил
sns.set(style="whitegrid", font_scale=1.1)

# Путь к результатам MapReduce
base_dir = os.path.expanduser("~/hadoop-lab-5/output_price_by_area")
result_path = os.path.join(base_dir, "part-00000")

# Загружаем результат: район, количество объявлений, средняя цена
df = pd.read_csv(
    result_path,
    sep="\t",
    header=None,
    names=["neighbourhood_group", "listing_count", "avg_price"]
)

display(df)

# --- График 1. Средняя цена ---
order_price = df.sort_values("avg_price", ascending=False)["neighbourhood_group"]

plt.figure(figsize=(8, 5))
ax1 = sns.barplot(
    data=df,
    x="neighbourhood_group",
    y="avg_price",
    order=order_price
)
ax1.set_xlabel("Район Нью-Йорка")
ax1.set_ylabel("Средняя цена, $")
ax1.set_title("Средняя цена аренды Airbnb по районам Нью-Йорка")

```

```

for p in ax1.patches:
    height = p.get_height()
    ax1.annotate(
        f"{height:.0f}",
        (p.get_x() + p.get_width() / 2, height),
        ha="center",
        va="bottom",
        fontsize=9,
        xytext=(0, 3),
        textcoords="offset points"
    )

plt.tight_layout()
plt.show()

# --- График 2. Количество объявлений ---
order_count = df.sort_values("listing_count", ascending=False)["neighbourhood_group"]

plt.figure(figsize=(8, 5))
ax2 = sns.barplot(
    data=df,
    x="neighbourhood_group",
    y="listing_count",
    order=order_count
)
ax2.set_xlabel("Район Нью-Йорка")
ax2.set_ylabel("Количество объявлений")
ax2.set_title("Количество объявлений Airbnb по районам Нью-Йорка")

for p in ax2.patches:
    height = p.get_height()
    ax2.annotate(
        f"{int(height):,}".replace(",", " "),
        (p.get_x() + p.get_width() / 2, height),
        ha="center",
        va="bottom",
        fontsize=9,

```

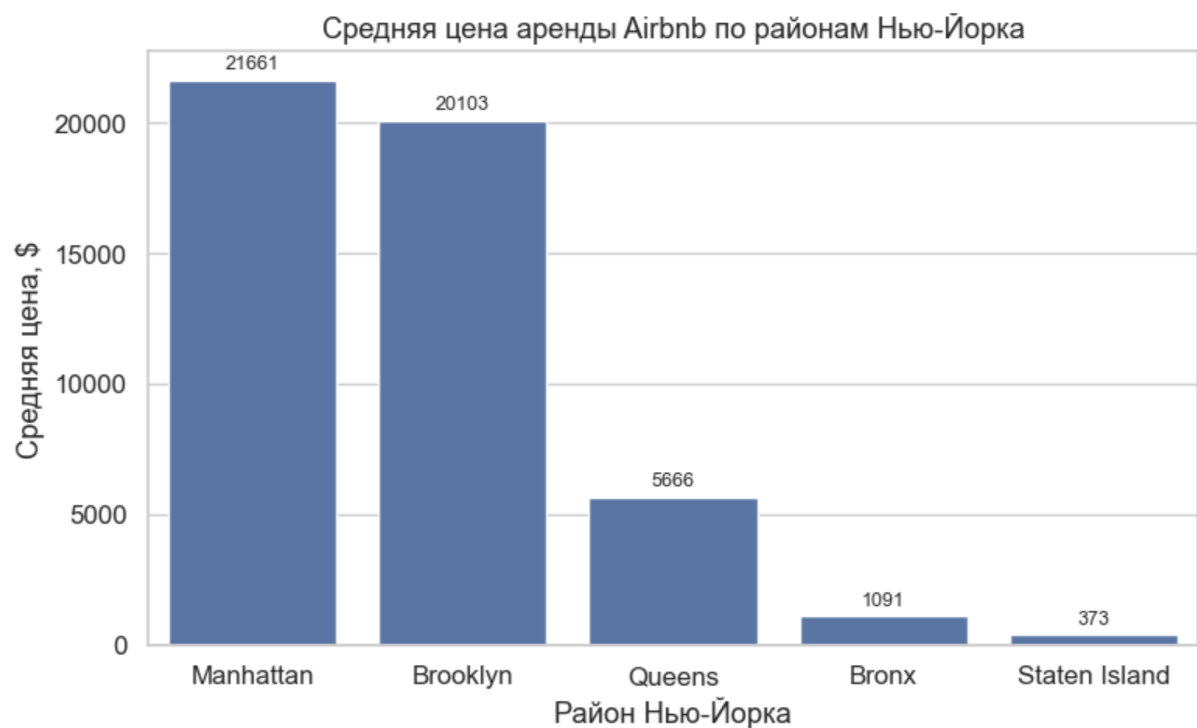
```

        fontsize=9,
        xytext=(0, 3),
        textcoords="offset points"
    )

plt.tight_layout()
plt.show()

```

	neighbourhood_group	listing_count	avg_price
0	Bronx	87.50	1091
1	Brooklyn	124.39	20103
2	Manhattan	196.88	21661
3	Queens	99.52	5666
4	Staten Island	114.81	373



Количество объявлений Airbnb по районам Нью-Йорка

